

Exercise 6: CLASSIFICATION - DECISION TREES

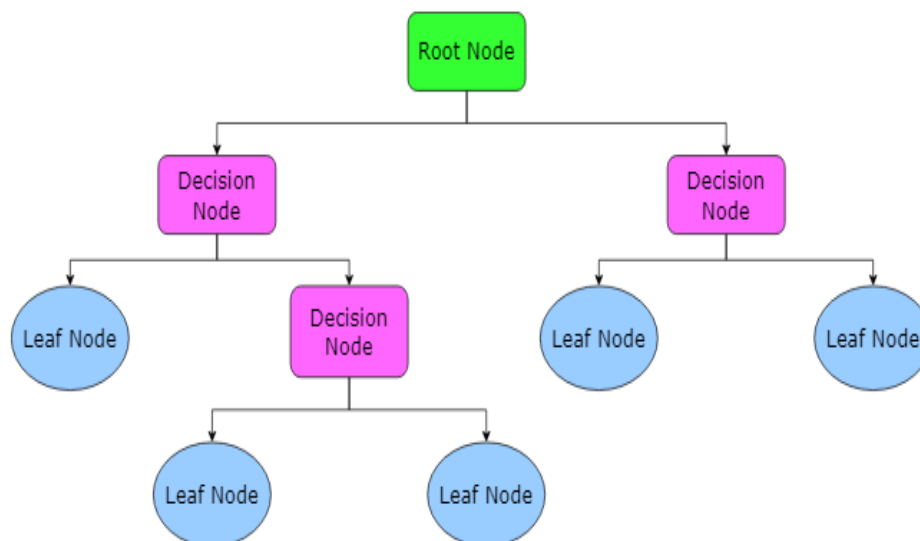
Write a python program

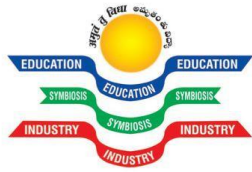
1. To build a decision tree classifier to determine the kind of flower by using given dimensions.
2. Training with various split measures (Gini index, Entropy and Information Gain)
3. Compare the accuracy

A **decision tree** is a machine learning algorithm that uses a tree-like model of decisions and their subsequent consequences to arrive at a particular decision. It is a *Supervised Machine Learning model*, where the data is continuously split according to a certain parameter, and finally, a decision is made.

Usually, a decision tree is drawn upside down, with the root node at the top and the leaf nodes at the bottom. A decision tree usually contains 3 types of nodes.

1. **Root node:** The very top node that represents the entire population or sample.
2. **Decision nodes:** Sub-nodes that split from the root node.
3. **Leaf nodes:** Nodes with no children, also known as **terminal** nodes.





Build a Decision Tree using IRIS dataset in Python:

```
import pandas as pd
import numpy as np
from sklearn.datasets import load_iris
from sklearn.metrics import accuracy_score

# Reading the Iris.csv file
data = load_iris()

# Extracting Attributes / Features
X = data.data

# Extracting Target / Class Labels
y = data.target

# Import Library for splitting data
from sklearn.model_selection import train_test_split

# Creating Train and Test datasets
X_train, X_test, y_train, y_test = train_test_split(X,y, random_state = 50, test_size = 0.25)

# Creating Decision Tree Classifier
from sklearn.tree import DecisionTreeClassifier
clf = DecisionTreeClassifier()
clf.fit(X_train,y_train)

# Predict Accuracy Score
y_pred = clf.predict(X_test)
print("Train data accuracy:",accuracy_score(y_true = y_train, y_pred=clf.predict(X_train)))
print("Test data accuracy:",accuracy_score(y_true = y_test, y_pred=y_pred))

import matplotlib.pyplot as plt
from sklearn.tree import plot_tree

plt.figure(figsize=(12,8))
plot_tree(clf, filled=True, feature_names=data.feature_names, class_names=data.target_names)
plt.title("Decision Tree Visualization")
plt.show()
```

Sample Decision Tree

